Question and design

Does model width improve contact accuracy at the same one-epoch exposure? Baseline: exp277, native plus MPNN corpus, 24 layers. Hidden width 2048 -> 4096; MLP width 8192 -> 16384. Keep 32 query / 8 KV heads; head width 64 -> 128. Keep data, optimizer schedule, seeds and global batch fixed.

Architecture verified

Actual Levanter parameter-tree counts: Baseline: 1,471,374,336 parameters. Wider model: 5,861,984,256 parameters (3.984x). 11 configuration and finite-epoch dataset tests passed. Count reproduction: count_parameters.py via the README command.

Current status

A 128-H100 smoke is being prepared in CoreWeave US-EAST-02A. 248 available H100s observed at 2026-10-13 14:43 UTC. Production follows only after the ten-update smoke succeeds. No training or contact-accuracy results yet.